

SAM3 prompting ablation, test split (source: recomputed; P/R at the F1-max confidence, IoU/Dice over matched instances)

SAM3 prompt condition	mAP@0.5	mAP@0.5:0.95	P	R	IoU	Dice	what it tests
SAM3 (text prompt, zero-shot)	0.028	0.020	0.081	0.141	0.840	0.909	class-name text prompts: detect + classify + segment, zero-shot
SAM3 (YOLO boxes)	0.499	0.345	0.655	0.500	0.809	0.887	YOLO11n boxes (+class, +score) as prompts: SAM3 only draws the mask
SAM3 (GT boxes, oracle)	0.970	0.669	0.977	0.977	0.831	0.904	GT boxes as prompts: perfect localisation + class = upper bound
YOLO11n-seg	0.543	0.379	0.696	0.527	0.810	0.889	reference: the detector whose boxes feed the row above